Assignment: Task Management System with Role-Based Permissions

Overview

Build a **Task Management System** where:

- Users can create **Projects**.
- Each project can have multiple **Tasks**.
- Tasks can be assigned to different users.
- Projects and tasks should support role-based permissions (e.g., Admin, Member, Viewer).
- Implement token-based authentication using **JWT**.

Requirements

1. **Models**:
    - **User**: Use a custom user model (inheriting from `AbstractBaseUser` and `PermissionsMixin`) with fields like `email`, `username`, `first_name`, `last_name`, and `is_verified`.
    - **Project**: A project model with fields like `title`, `description`, `created_at`, `updated_at`, and a relationship to `User` (creator).
    - **Task**: A task model with fields like `title`, `description`, `status` (e.g., "To Do", "In Progress", "Completed"), `assigned_to`, and a foreign key to `Project`.
    - **ProjectRoles**: A model to define roles for users in a project (e.g., Admin, Member, Viewer).
2. **API Endpoints**:
    - **Authentication**:
        - Register and login users using JWT authentication.
        - Email verification during registration.
    - **Project Management**:
        - Create, update, delete, and retrieve projects.
        - Manage roles for users in a project (assign/remove roles).
    - **Task Management**:
        - CRUD operations on tasks within a project.
        - Assign/unassign tasks to users.
3. **Permissions**:
    - Only **Admins** can manage roles and tasks in a project.
    - **Members** can update task statuses.
    - **Viewers** can only view project and task details.
4. **Additional Features**:
    - Use **filters** (e.g., by status, assigned user) and **pagination** for task lists.
    - Implement a search feature to find tasks by title or description.

Bonus Points:

1. Implement WebSocket notifications using **Django Channels** when a task's status is updated.
2. Add a **Swagger/OpenAPI** documentation for all APIs.
3. Use **Celery** to send an email notification when a task is assigned.

Deliverables:

1. A GitHub repository with clear commit messages.
2. A `README.md` file with instructions to set up and run the project locally.
3. Deployed application URL.